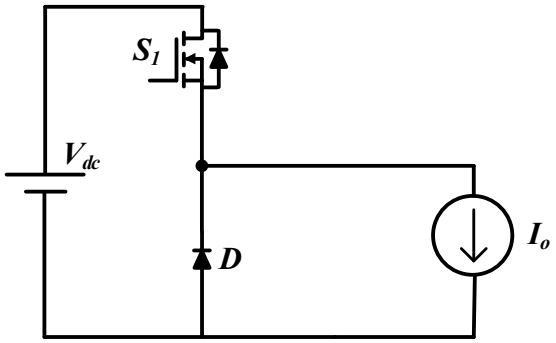
WEEK 3: ASSIGNMENT

1. While doing small signal modelling to obtain the transfer function, the perturbation occurs after

A. Average large signal model
B. Linearization around the operating point
C. Conversion from time domain to s-domain

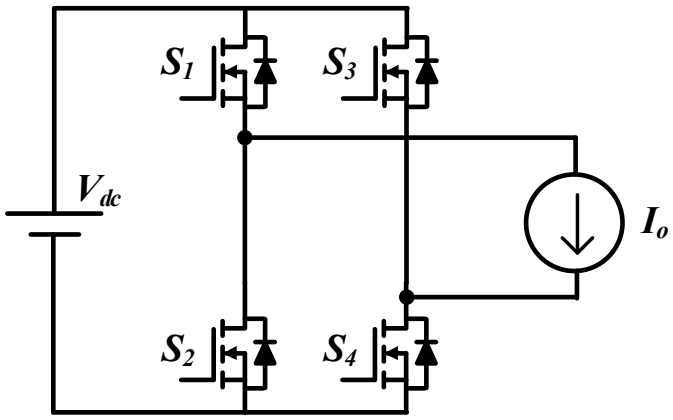
Answer: A

2. For the circuit shown below what will be the RMS current (in A) of the switch S_1, if the switch S_1 is turned-on for 30 µs, while the switching frequency of the switch is kept at 10kHz, $V_{dc} = 200V$, $I_o = 20A$. The answer should be rounded up to 3 decimal places. Assume the deadtime is negligible.



Answer: 10.500 to 11.500

3. For the circuit shown below what will be the RMS current (in A) of the switch S_1, if the switches S_1 and S_4 are turned-on at the same time for 30 µs, while S_2 and S_3 are turned-on at the same time for 20 µs. S_1 and S_2 are switched in complementary manner; S_3 and S_4 are switched in complementary manner. Assume $V_{dc} = 400V$, $I_o = 10A$ and deadtime is negligible. The answer should be rounded up to 3 decimal places.



Answer: 7.700 to 8.700

4. If a single-phase PFC converter switching at 10 kHz has perfect unity power factor current drawn is fed from 230V, 50Hz AC input, then output capacitor voltage ripple has following dominant frequency component

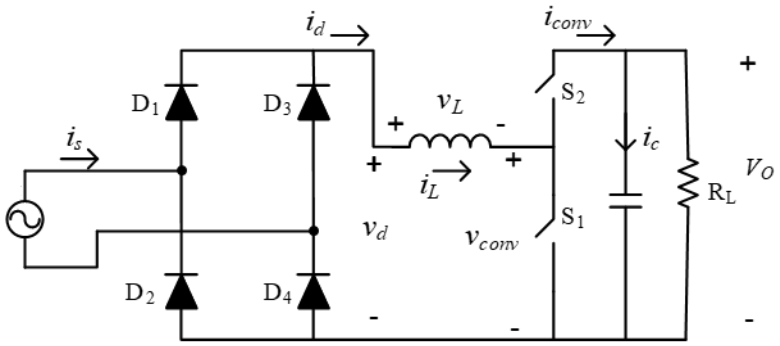
A. 50 Hz
B. 100 Hz
C. 10 kHz
D. 20 kHz

Answer: B

5. A boost PFC converter fed from 230V, 50Hz input is supplying the 3300W load. The converter is operated with the switching frequency of 20 kHz and regulates the output voltage at 400V. The output capacitance of 2.2 mF is kept at its output, then calculate the peak to peak output voltage ripple. (Note: The answer should be rounded up to 2 decimal places; use $\pi = 3.14$; $1\ radian = 57.3\ degrees$).

Answer: 11.50 to 12.50

6. . For the Boost PFC as shown below, what will be the expression of RMS current of switch S_2



A. $\sqrt{\frac{2}{T}\int_0^{\frac{T}{2}} \mathrm{d}(t)(i_s^2(t))\,dt}$

B. $\sqrt{\frac{2}{T}\int_0^{\frac{T}{2}} (1-\mathrm{d}(t))(i_s^2(t))\,dt}$

C. $\frac{I_{s,pk}}{2}$

D. $\frac{I_{s,pk}}{\sqrt{2}}$

Answer: B

7. For the table shown below:

Matrices	Matrix composition
I. A	a. 2 X 2
II. B	b. 1 X 2
III. C	c. 2 X 1

Match the correct combination (Symbols have obvious meanings):

A. I -a, II-b, III-c
B. I -a, II-c, III-b
C. I -b, II-a, III-c
D. I -b, II-c, III-a
E. I -c, II-a, III-b
F. I -c, II-b, III-a

Answer: B

8. Which of the following is/are the correct assumptions which have been taken while obtaining the small signal modelling of the boost PFC converter to obtain the average current control?

A. $f_{sw} \gg f_s$
B. $f_s \gg f_{sw}$
C. magnitude of perturbations $\gg$ operating point values
D. magnitude of perturbations $\ll$ operating point values
Answer: D, A

9. Which of the following quantities is/are the state variables in the boost PFC converter?

A. Inductor current
B. Duty ratio
C. Input AC voltage
D. Output capacitor voltage

Answer: A, D

10. Which of the following quantities is/are the input variables in the state space representation of boost PFC converter?

A. Inductor current
B. Duty ratio
C. Input AC voltage
D. Output capacitor voltage

Answer: B, C

Answer Key:

1. A	2. 10.5 to 11.5
3. 7.7 to 8.7	4. B
5. 11.5 to 12.5	6. B
7. B	8. D, A
9. A, D	10. B, C